MPRA Benchmark Paper Meeting Notes

Idea

Benchmark the top DNA-sequence models (e.g., Enformer, AlphaGenome) against your MPRA data by comparing each model's predicted activity change for reference vs. alternate alleles to the experimentally measured MPRA effects, then rank performance and prepare reproducible code and figures for publication.

Model Selection

Model	Names
Enformer	Musti
Basenji2	Arjun
AlphaGenome	Kilian
Borzoi	(not needed maybe)
DeepSEA	(not needed maybe)
PromotorAI	
DNABert	Kilian (NT)
HyanDNA	Musti
Evo2	

Planning

- Catalogue every MPRA library
    - Sequence, tissues, replicates
- Compute activity in a standardized manner
- Prediction generation
    - Extract model windows around each tested allele
    - Infer reference and alternate sequence
    - Derive unified Activity metric
- Benchmarking

Meeting Notes:

04.08.25
Attendees: Martin, Musti

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Model Selection (by class)

- Genome-wide multi-track predictors (supervised): Enformer, Basenji2, Borzoi
- Variant scorers: DeepSEA/Beluga, AlphaGenome (variant head)
- Foundation LMs (LLMs / smaller Transformers): DNABERT-2, Nucleotide Transformer (NT), HyenaDNA
- Specialized (promoter): PromoterAI (baseline)
- Wrappers/Hubs: Hugging Face, Kipoi (for reproducible inference)

What exactly do we compare? (Martin's focus)

- Element vs Variant benchmarks: run both tracks.
- Element: predict element activity (one score per oligo/tile).
- Variant: predict allele-specific effect Δ(ALT−REF) within the oligo.
- Cell-type vs Organism-level: report cell-matched where available; also a tissue-agnostic (organism-level) aggregation as a secondary view.
- Outputs & tracks: pre-map assay to biology
- Enhancer MPRA → ATAC/DNase (accessibility)
- Promoter MPRA → CAGE/RNA (expression)
- Multi-output handling: models differ in outputs/positions → use predeclared reduction over the oligo span (primary: mean; robustness: sum & top-k mean). For profile/count outputs, apply log1p/asinh first.
- Window vs. oligo: run a fixed kb input window, but compute scores only within oligo bounds. If the window extends beyond oligo: either pad with N's or also score endogenous locus and report both.

Normalization & Targets

- MPRA: compute effect sizes; for cross-dataset comparability, z-score within each library.
- Note (Martin): Element activity need not be normalized for within-library classification; we'll still provide z-scores for cross-library regression.

Metrics (what do we correlate with what?)

- Primary (regression): Spearman between model score and MPRA effect.
- Secondary (classification): AUROC/AUPRC for activity/effect thresholds (significance defined by barcode coverage + sequence design).
- Directional accuracy (sign agreement), Pearson r, and calibration slope.
- Power-aware: weight or filter by replicate SE / barcode coverage (handles small-effect/low-coverage vs high-effect/high-variance).

14.08.25
Attendees: Max, Kilian, Arjun, Musti

RECAP
Benchmark leading DNA sequence models on MPRA-measured regulatory variant effects, with standardized processing, fair mapping from model outputs to MPRA readouts, and reproducible leaderboards.

- Element vs variant, c-type vs organism-level,plasmid vs lenti assays.
- Suggested Moldes: Genome-wide sequence to signal

Positioning & Related Work
- ProxReg will be a resource paper (data + benchmark scaffolding; no new biology).
- Inspiration: Max's paper + guanine repo → consider a "non-coding variant gym"–style benchmark hub (harmonized MPRA + standardized model eval).
- Keep variant-level as immediate focus; element-level also included.
- Compare models on elements too: which models suit promoters vs enhancers and common genomic region

Evaluation & thresholds
- Primary metrics: Spearman regression and AUROC (classification).
- Effect bins: define low / high effect AUROC/AUPRC per bin.
- Significance policy: label positives by barcode-aware significance (coverage + design) maintain per-library thresholds and document them.
- Expect better predictability farther from 0 → always stratify by effect size.
- Minimum barcode coverage filters; report coverage distributions.
- Handle small-effect/low-coverage vs high-effect/high-variance explicitly in stratified results.
- Build promoter-like and enhancer-like subsets (annotation-driven)

Benchmark axes
- Element vs Variant (run both).
- Cell-type matched vs organism-level (tissue-agnostic mean).
- Assay class: plasmid (Tewhey) vs lenti (our data)
- Distance to TSS: expect stronger performance near promoters; test toward/away from TSS.

How do we handle window vs. oligo ?
- Run all models on a fixed kb window (e.g., 1 kb).
- Reduce only within the oligo bounds (primary: mean; robustness: sum and top-k mean).
- If needed, pad with N's to fill the window; document vector effects if leaving oligo span.

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What baseline do we include ?

- Simple: GC content, k-mer LR, PWM motif disruption score, ATAC overlap (if available)

Next steps

- Model shortlist (confirm): Enformer, Basenji2, Borzoi, DeepSEA/Beluga, AlphaGenome (variant), DNABERT-2, NT; (stretch) HyenaDNA, PromoterAI baseline.
- Workflow: containerized inference wrappers → standardized reduction → eval; start with ProxReg pilot
- Meetings: tools discussion 18/08, then biweekly Thu 13:00.
- Set up a Git repo
- Organize the data that we will be using.
- Split models to test between people

08.09.25

Attendees: Martin, Kilian, Arjun, Musti

Inputs for element analysis

- Region tested: default = centered on element midpoint.
- Sensitivity: also test start and end anchored windows.
- Include PromoterAI to study promoter distance effects on scores

Model-specific notes & actions

Arjun — Basenji2

- Variant scoring implemented; predicts many tracks (128 bp resolution), ~100 kb input.
- Decide MPRA proxy: ATAC/DNase for enhancer MPRAs; CAGE for promoter MPRAs.
- Optional: max across TF tracks as a secondary feature.
- Actions: set up Basenji2 on BIH, validate VCF-based scoring, finalize track list.

Kilian — AlphaGenome

- Outputs: RNA-seq, CAGE, DNase at bp resolution; TF binding at 128 bp.
- Min input ~2 kb (sequence length parameter).
- Needs VCF input; confirm exact CLI/API and windowing.
- Actions: finalize VCF workflow; pick primary tracks per assay.

Musti — Enformer

- Action: review weights, inference wrapper, and track selection; confirm reduction path (profile → log1p/asinh → mean/sum/top-k within oligo

To Do List

- [ ] Confirm core model list + track mapping (per library).
- [ ] Basenji2: BIH setup, VCF pipeline, primary/secondary track reductions.
- [ ] AlphaGenome: VCF workflow, min-length param, track choices.
- [ ] Enformer: wrapper + weights check; reduction spec finalized.

- [x] ~~Element input centering default; start/end sensitivity runs scripted.~~
- [x] ~~Implement mean / sum / |sum| reductions; add effect-size stratification in eval.~~
- [x] ~~Repo + OMICS scaffolds; auto-figure generation in pipeline.~~

Enformer Research

- Enformer supports variant-effect scoring via SNP Activity Difference (SAD = ALT − REF) across many tracks (CAGE, DNase/ATAC, histone)
- precomputed SAD scores for common variants are available to sanity-check
- Colab Code Available for Variant Scoring - enformer-usage.ipynb - Colab
- Input: 200kb sequence, Output hundreds of tracks on 128 bp grids
- Kiopi Variant Scoring Option: Variant effect prediction - Kipoi-veff
    - it's a **generic variant-effect scoring interface** one can use it to *standardize* how you score REF vs ALT for **any** sequence model
- Variant effect prediction of promoter variants using the Enformer model gagneurlab/AbExp-utils

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23.09.
Attendees: Arjun, Musti

Arjun / Basenji2 status
- Ran Basenji2 on a subset of variants; has initial plots.
- VCFs generated per experiment / per cell line for ProxReg.
- CPU cluster not viable (too slow) → CPU bottleneck for full run.

Data
- VCF per experiment/cell line (ProxReg).
- Need a smaller VCF for testing runs
    - Use to stabilize model wrappers and compare models consistently.
- Arjun will upload VCFs + plots to the repo and add a small test set.

Variant scoring (SAD)
- Basenji2 provides SAD-like variant scores (ALT−REF across tracks).
- Enformer also supports SAD.
- Plan: interpret outputs on the test VCF, then compare Basenji2 SAD vs Enformer SAD on the same subset.

Practical constraints
- Prefer GPU runs for Basenji2; avoid large CPU jobs.
- Keep VCFs small & targeted for quick iteration; scale after validation.

Next steps
- [x] ~~Arjun: upload VCFs, plots, and test VCF.~~
- [x] ~~Understand output of the tools to harmonize~~
- [x] ~~Test runs with subset of variants VCF~~

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06.10.2025

Participants: Kilian, Arjun, Musti

Input structure:
- Variant specific: vcf (like
- Element specific: (p
- robably bed but not important now)

Output-structure:
- tsv
- Expected column names: variant name, score column(s), measured variant score
- score column: <model_names>.<comparison_type>.variant_score.<optional_info>
- combined variant score results:
    - basenji (default deltas)
        - activity score (all tracks) (arjunvdevadas@gmail.com can you give more details on how the activity score is computed?)
        - expression difference
        - mean score
        - max score
    - enformer (default deltas):
        - difference between reference and alternative + sum
    - alphaGenome (what are the default deltas):
- Combination type of a score: (was e.g. mean or max)
    - “abs_mean”, “abs_max”, “not_abs_mean”, “not_abs_max”

Next steps

- [ ] Pushing the code for the variant effects
- [x] ~~Every model has an explainable model~~
- [x] ~~Explainable Score~~
- [x] ~~3+ Column Output~~
- [ ] Try out some figure
- [ ] Github - Own Branch per Model
- [x] ~~Add Data Folderstructure~~

20.10.2025

Participants: Kilian, Arjun, Musti

- How to harmonize data – main goal of this talk
- Find intersection of predictors between tools
- basenji provides one final score per assay
- We have assay data for AlphaGenome and Enformer
- Basenji is target_list based, so it should be possible
- Paper should also compare tool performance between each other
- get values for assay specific data
    - Mean for Assay type
    - Maybe the difference between assay types
- AlphaGenome has ATAC instead of DNase

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- List of all the Features / Targets for each model
- PROJECT: How to best aggregate all assay data into one unified score
- VCF file has all information of MPRA data already
- Merge all tables at the end

To-Do

- [x] ~~Average of the assay data per tool~~
    - Enformer, Basenji (cage, chip, dnase)
    - Alphagenome (ATAC, Chip, …)
- [x] ~~have a exe with VCF as Input~~
    - Also have a env.yaml
- [x] ~~Hand Over Basenji to Musti~~
- [x] ~~Move Arjun project folder to Project Folder~~
- [x] ~~Push Code To Github~~
- [ ] Get a list of Top Aligning Variants (Top 10% or top cor. of > 0.95)
    - Think about on how to compare these in a good way

Musti 30.10.25

- Im Adding Basenji SAD into the Pipeline
- Pushed Pipeline
- Visualizations and Metrics are running

17.11.2025
Participants: Kilian, Martin, Musti

- Discussion of wider model list
    - different architectures
    - sequence to function modelle
- Padding vs no Padding
- Variant Interpretation model vs mpra because of context window is a problem
    - only 270bp in model
    - binning of regions needs to be performed somehow
    - only effect of mpra size
- Center of Variant - MPRA is not always centered
    - maybe sub analysis, if positions have an effect
- Categories of Variants
- different AF, eQTLs, needs to be diverse

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Next steps

- [ ] Add more models to test on (min 7)
    - [ ] min one LLM
    - [ ] Hyna (Musti), DNABert, NT (Killian)
- [ ] New data, Ryan Data from Encode maybe
- [ ] Kilians data, Nicks Data
- [ ] Meta Data Question, what is the main influencing factor ?
    - How regions were sampled, how variants were selected, MPRA Sequence length, and so on
    - Standing variations vs. non-Standing
      Q: How do we calculate Scores from model Tracks?
      Q: Is MPRA actually a good standard ?
- [ ] Maybe Say MPRA is state of the Art because of ENCODE, just analyse models ?
- [ ] If every model predictions scoreboard change with different cell type, is my data actually correct ?
- [ ] maybe make hepG2 broder - take everything that is Liver ?
- [ ] Change Metrics around
- [ ] ref vs alt oder nucleotide dependencies ?
- [ ] NT2, Hyan, Evo2,
- [ ] Push Code again
- [ ] Read Me for data Merge Table for vis
- [ ] outline for manuscript
    - [ ] what questions do we wanna answer
- [ ] NT2 or DNABert

Documents:

- https://pmc.ncbi.nlm.nih.gov/articles/PMC12340898/
- Take Paper as inspo for Model Selection